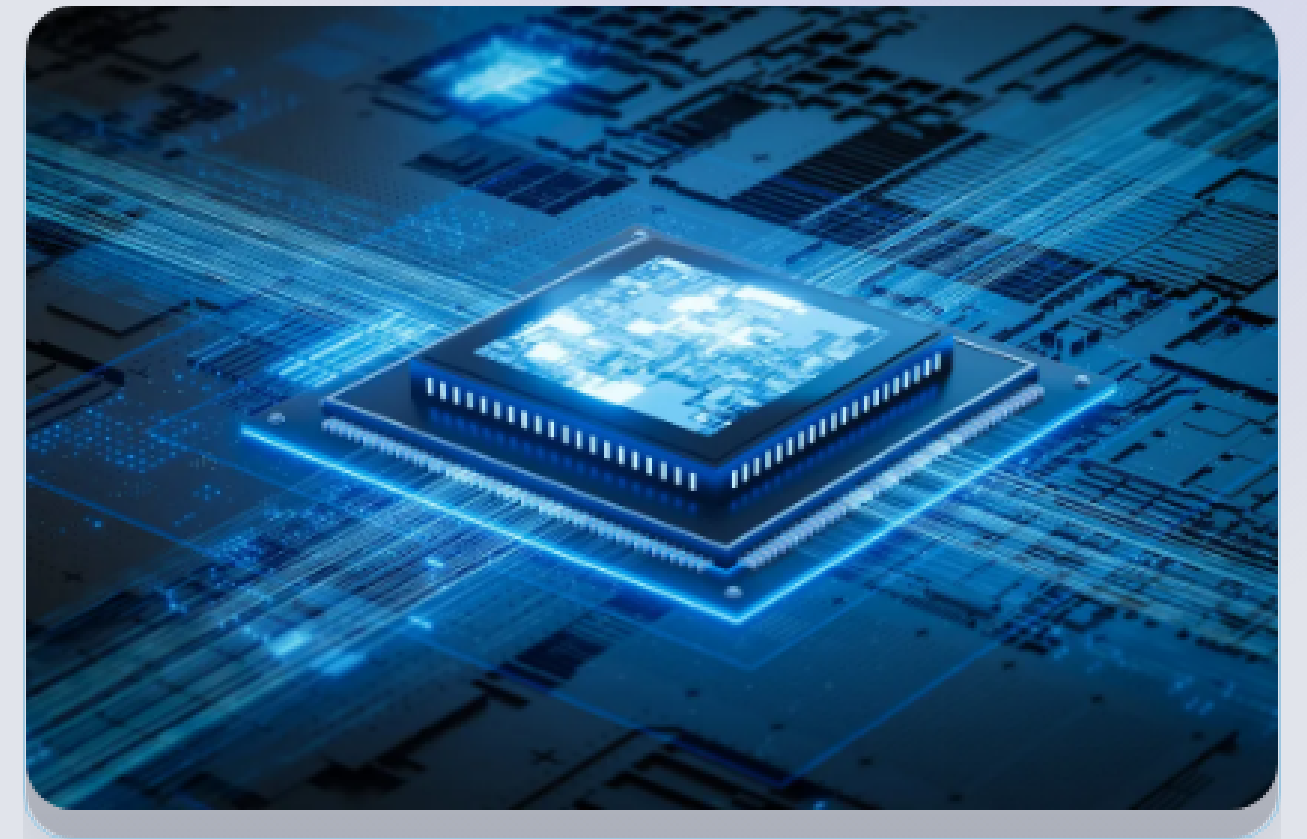


Neural Processing Unit (NPU) & SoC: System on Chip

Next-Generation Hardware Acceleration & Deep Learning System Architecture



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Neural Processing Unit (NPU)

An overview of NPU architecture, core capabilities, key advantages over traditional CPUs, and real-world AI applications.

What is a Neural Processing Unit (NPU)?

Specialized AI Accelerator

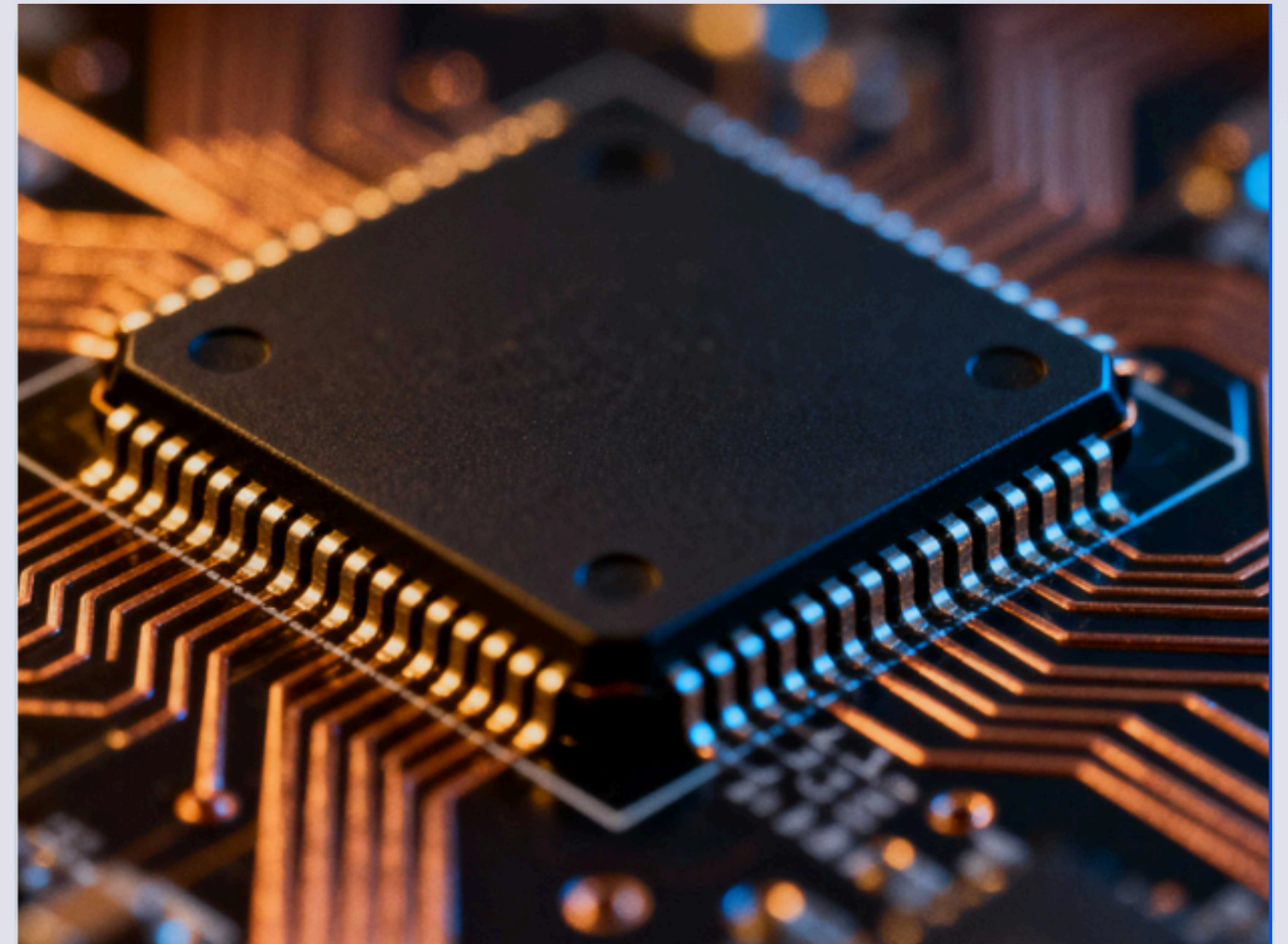
An NPU is a dedicated microprocessor designed specifically to accelerate Artificial Intelligence (AI) and Machine Learning (ML) tasks efficiently.

Matrix & Tensor Optimization

Unlike general-purpose processors, NPUs are engineered specifically for mathematical operations essential to neural networks, such as complex matrix calculations.

System Integration

NPUs are commonly integrated directly into modern System-on-Chips (SoCs) alongside CPUs and GPUs to power intelligent features seamlessly.



NPU vs. CPU: Architectural Comparison

Feature / Metric	Central Processing Unit (CPU)	Neural Processing Unit (NPU)
Primary Focus	General-purpose computing tasks	Specialized AI & ML computations
Execution Style	Sequential execution / Few complex cores	Massively parallel / Tensor matrix engines
AI Efficiency	Lower efficiency for neural networks	Ultra-high throughput & optimized performance
Power Consumption	Higher energy draw under heavy AI load	Low power draw optimized for battery devices
Primary Workloads	OS operations, logic, web browsing, apps	Facial recognition, speech, generative AI

Key Features & Technical Advantages



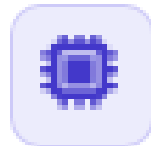
Optimized for AI

Designed with dedicated hardware architecture specifically tuned for neural network algorithms, deep learning layers, and tensor operations.



Parallel Processing

Executes thousands of matrix math calculations simultaneously, providing the extreme data throughput essential for real-time AI workloads.



Low Power Consumption

Delivers significantly higher performance-per-watt, dramatically saving battery life in mobile devices like smartphones and laptops.



On-Device Privacy

Enables local AI processing directly on the hardware. Reduces latency and keeps sensitive personal data on the device without cloud dependency.

Real-World Applications of NPUs

Smartphones

Powers biometric face recognition, computational camera enhancements, scene detection, and virtual assistants.

Computer Vision

Enables real-time visual analysis, object detection, spatial tracking, and automated image segmentation.

Voice Recognition

Accelerates on-device speech-to-text conversion, real-time audio translation, and noise suppression.

Generative AI

Runs local Large Language Models (LLMs) and image generation tools directly on user hardware.

Autonomous Vehicles

Processes real-time camera feeds and sensor arrays for instantaneous intelligent driving decisions.

Security Systems

Drives smart surveillance systems with real-time video analytics and instant threat detection.



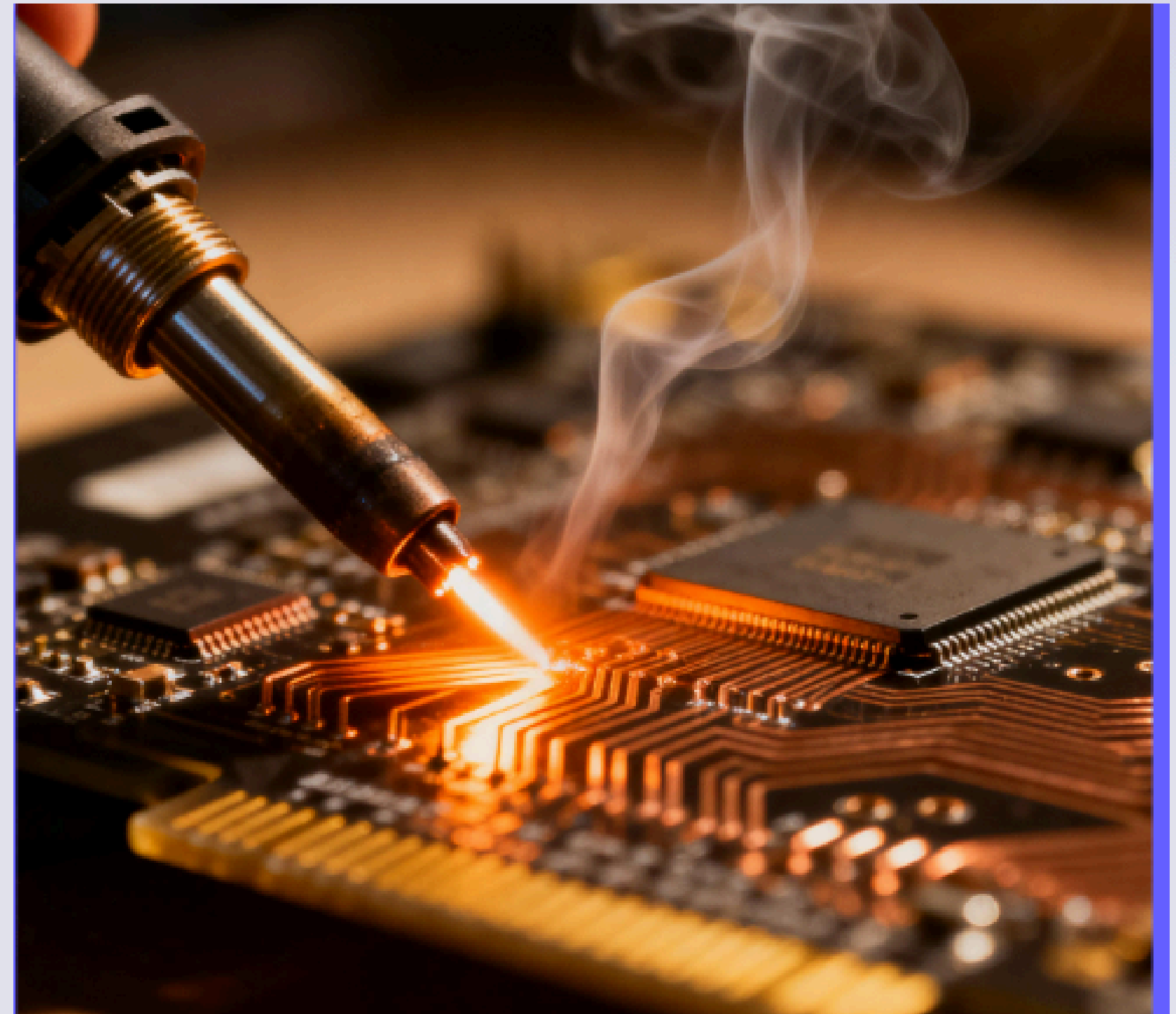
SoC Integration Note: NPUs rarely operate in isolation. They are integrated alongside CPUs and GPUs into a single chip known as a **System on Chip (SoC)**.

⚙️ NEXT-GEN HARDWARE ARCHITECTURE

SoC: System on Chip

Integrating CPUs, NPUs, and System Controllers into a single, high-efficiency silicon platform for modern AI workloads.

📁 Architecture Overview & NPU Integration



What

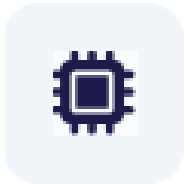
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System on Chip

SYSTEM ON CHIP

A **System on Chip (SoC)** is an integrated circuit that combines most of the major components required to perform a computing system's functions onto a single chip.

SoC can contain :



CPU

Central Processing



ISP

Image Signal



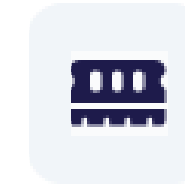
NPU

Neural Processing



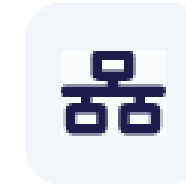
GPU

Graphics Engine



Memory

RAM Controller

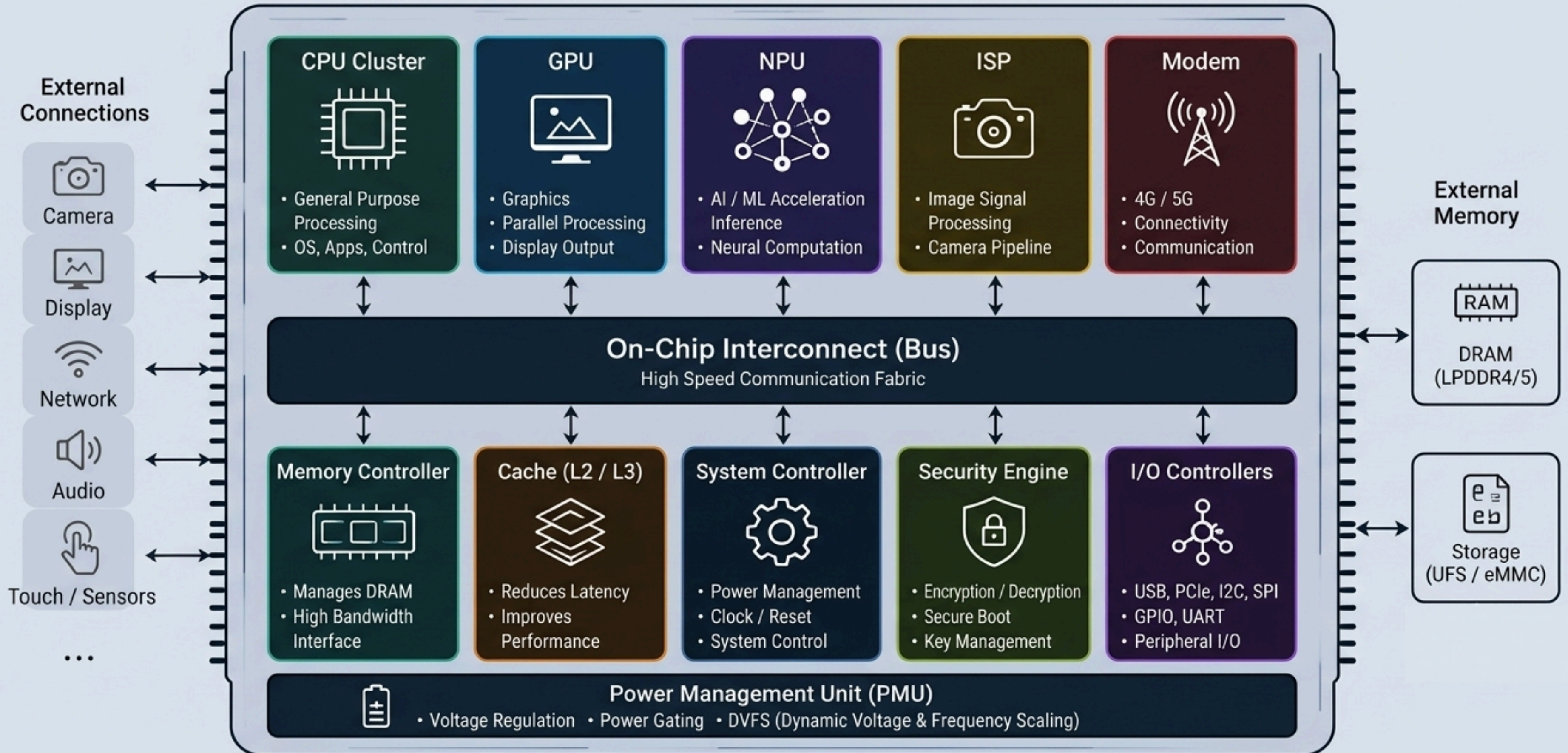


Bus / I/O

Peripherals

SoC (System on Chip) Architecture

All major components integrated on a single chip



Core Modules within an SoC

CPU Core

Handles general-purpose computing tasks, executing OS instructions and sequential workload logic.

GPU Unit

Accelerates complex 2D/3D graphics rendering, video decoding, and parallel math processing.

NPU Accelerator

Dedicated neural engine optimized for low-power machine learning models and AI inferencing tasks.

ISP Module

Processes raw camera sensor data in real-time, handling auto-focus, noise reduction, and HDR imaging.

Memory Controller

Manages high-speed data flow between system caches and unified system RAM (LPDDR memory).

Security & I/O

Houses hardware security enclaves, cryptographic engines, and external bus interfaces.

**BUT
WHY?**



Why Integrate Everything Into One Chip?

Efficiency



Ultra-Low Power Consumption

Shorter physical wire traces eliminate signal resistance and reduce dynamic power dissipation across motherboard traces.



Sub-nanosecond Latency

High-speed on-chip interconnect buses allow CPU, GPU, and NPU to share data near-instantly without memory bottlenecking.



Miniaturized Form Factor

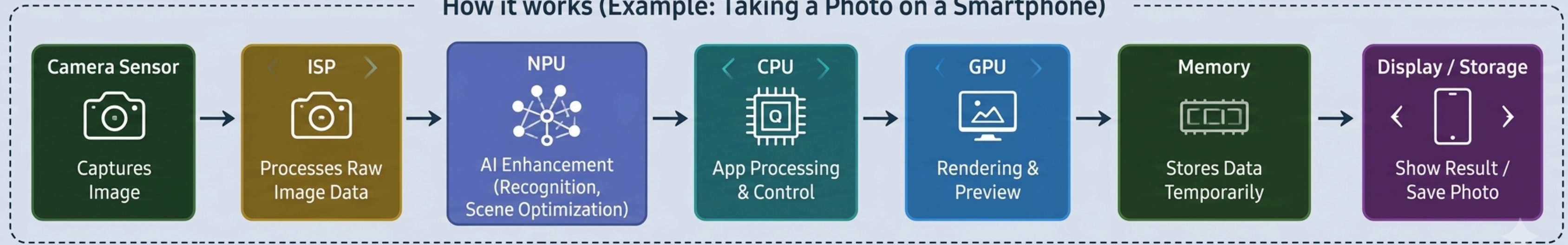
Combining multiple chips into a single package saves critical board space in smartphones, wearables, and thin laptops.



Cost-Effective Mass Production

Lower overall bill-of-materials (BOM) cost by reducing discrete component assembly, testing, and packaging steps.

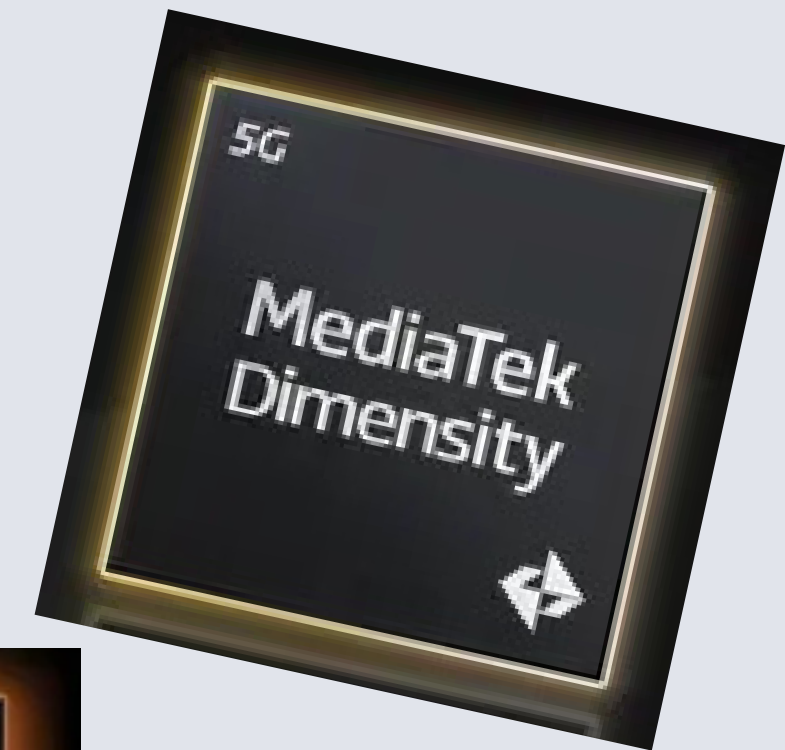
How it works (Example: Taking a Photo on a Smartphone)



Key Idea: Different specialized units work together through high-speed on-chip interconnect and shared memory to deliver high performance with low power in a compact form factor.

ARM architectures define the mobile SoC landscape

In smartphones, chips based on the ARM architecture are all built on this SoC concept.



Desktop CPU Evolution & APU



Integration Evolution

- > Core-Only Era: Early Intel and AMD processors provided only CPU processing cores on the die.
- > On-Die Cache: Intel pioneered adding high-speed Cache memory directly into the CPU chip.
- > Integrated Memory Controller: AMD moved the memory controller off the motherboard directly onto the CPU silicon.
- > Core Benefit: Dramatically increased output, boosted overall system efficiency, and significantly reduced memory access latency.

Cache Integration

Lower Latency



APU & Integrated GPUs

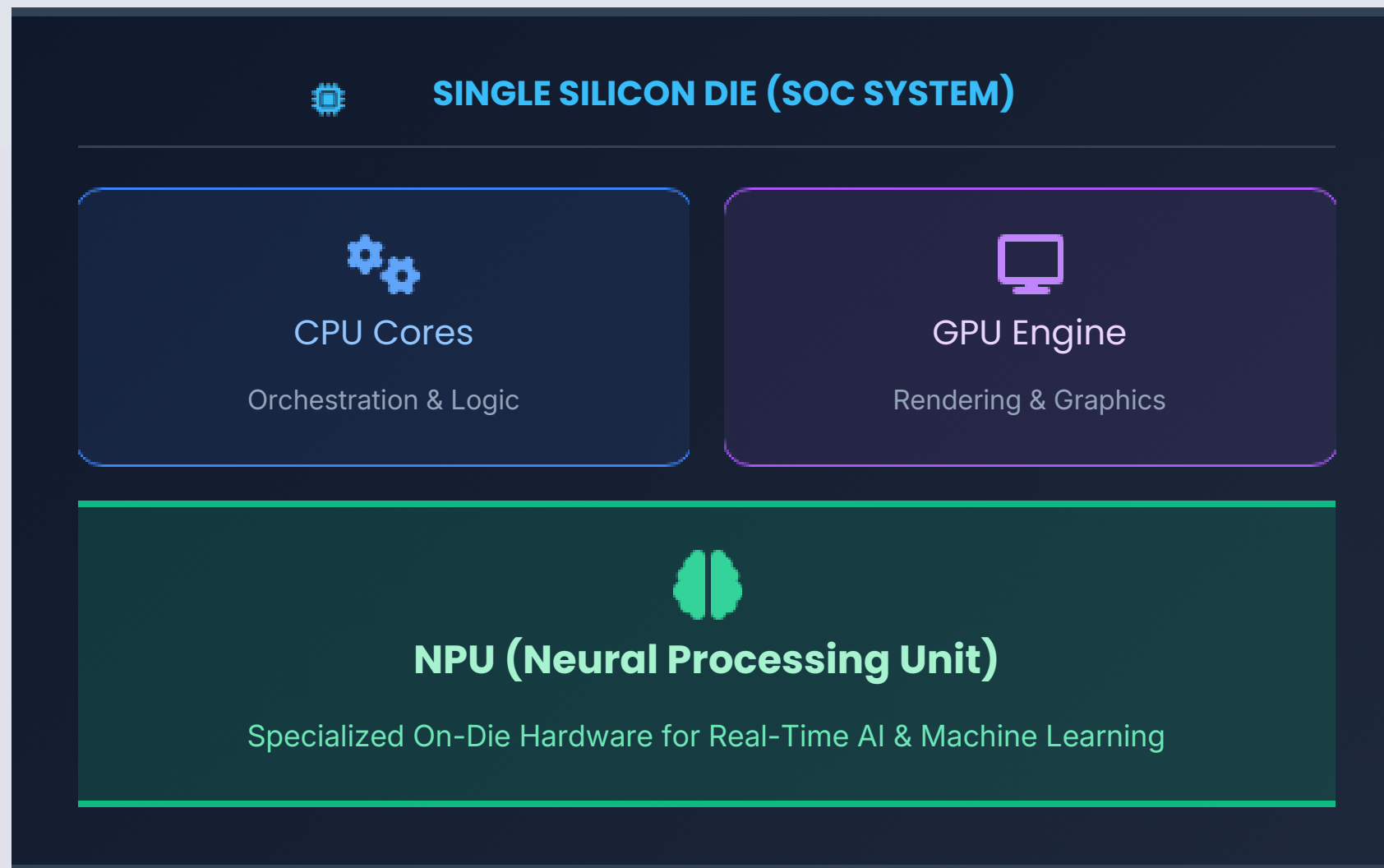
- > Unified Chip Design: APUs (Accelerated Processing Units) and iGPUs integrate both CPU and GPU computational cores onto a single silicon die.
- > Workload Versatility: Excellent performance for low to medium-level graphics tasks without requiring a dedicated discrete graphics card.
- > Key Applications: Video editing, file compression, mid-level gaming, and light numerical simulations.
- > Space & Power Saver: Reduces overall power draw, heat generation, and physical space within computing hardware.

CPU + GPU Single Die

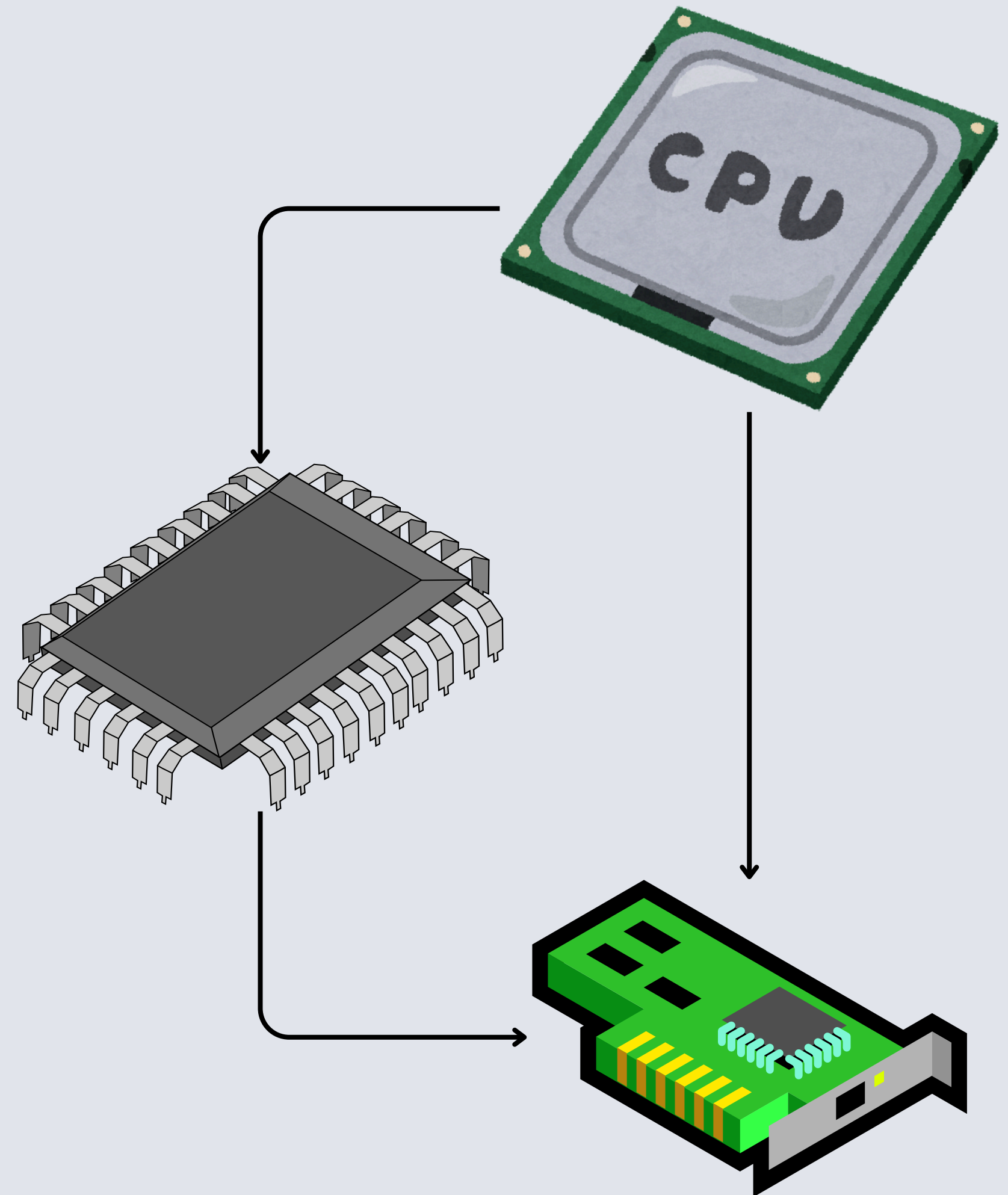
Medium Graphics

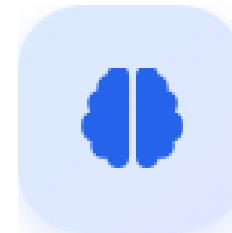
Efficient Power

NPU Integration inside the SoC



An **NPU does not have to be a separate chip**. It operates directly inside the SoC alongside the CPU & GPU as one unified, high-efficiency system.





Thank You Everyone for Your **Attention**



"Attention Is All You Need"

— Vaswani et al. (Google Research, 2017)



Q&A & Discussion



System-on-Chip & NPU Architecture